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Department of computer science and engineering

MAJOR PROJECT PHASE-I - 22CSP69

SLAM NAV – Low-Cost Monocular Visual SLAM Using a Monochromatic Camera

Presentation By:

Anish Bharadwaj – 1NT23CS024

Dhanush Paturu – 1NT23CS060

Dhruva PS – 1NT23CS064

Guided By:

Dr. Venkatesh Krishna Rao

Designation - Professor

AY 2025-2026

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Background

- Simultaneous Localization and Mapping (SLAM) is a fundamental problem in robotics and autonomous systems.
- SLAM enables a robot or device to **estimate its position while simultaneously building a map of the environment**.
- Traditional autonomous navigation systems often rely on **LiDAR sensors** for accurate depth and mapping.
- However, LiDAR systems are **expensive, power-intensive, and bulky**, making them less suitable for lightweight or low-cost robotic systems.
- Vision-based SLAM has emerged as a **cost-effective alternative**, using cameras instead of specialized sensors.
- In this project, we explore a **monocular SLAM system using a monochromatic camera** to perform localization and mapping efficiently.

Research Motivation

- Many robotics applications require **low-cost and lightweight navigation systems**.
- LiDAR-based SLAM systems significantly increase the **cost and hardware complexity**.
- Monochromatic cameras are **cheap, compact, and widely available**, making them attractive for robotics applications.
- Monochromatic cameras also offer **better light sensitivity and improved edge detection** compared to RGB cameras.

Problem Statement

- Existing monocular SLAM systems often suffer from:
  - **scale ambiguity**
  - **tracking failures in low-texture environments**
  - **drift over long trajectories**
- Therefore, there is a need to design a **robust monocular SLAM system that can perform localization and mapping using only a single monochromatic camera**.

Literature Survey

- Chen, M., Guo, H., Qian, R., Gong, G., and Cheng, H.: Visual simultaneous localization and mapping (vSLAM) algorithm based on improved Vision Transformer semantic segmentation in dynamic scenes, Mech. Sci., 15, 1–16, https://doi.org/10.5194/ms-15-1-2024, 2024.
- B. Fu and X. Yao, "Visual SLAM System Integrating Feature Extraction Based on Vision Transformer and Cloud-Edge Technology," 2024 International Conference on Intelligent Computing and Robotics (ICICR), Dalian, China, 2024, pp. 73-77, doi: 10.1109/ICICR61203.2024.00023. keywords: {Location awareness;Deep learning;Visualization;Simultaneous localization and mapping;Accuracy;Feature extraction;Transformers;Transformer;Cloud-edge technology;Feature Extraction;Deep Learning;Visual SLAM},
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- S. Schubert, “On the potential of visual place recognition for visual SLAM,” Chemnitz University of Technology, Germany.
- S. Izquierdo and J. Civera, “Optimal transport aggregation for visual place recognition,” University of Zaragoza, Spain. Available: https://github.com/serizba/salad.

GAP Analysis

- **LiDAR-based SLAM systems**, while highly accurate, are **expensive, bulky, and require significant computational resources**, limiting their adoption in low-cost robotic platforms.
- **Traditional visual SLAM methods** primarily rely on **handcrafted feature extraction techniques** such as ORB, SIFT, or FAST, which often struggle in **low-texture or low-light environments**.
- Existing research in visual SLAM has largely focused on **RGB or stereo camera systems**, with **limited exploration of monochromatic camera-based SLAM**, despite the advantages of monochrome sensors in terms of **light sensitivity and image clarity**.
- Although **modern deep learning techniques such as Convolutional Neural Networks (CNNs) and Vision Transformers (ViT)** have shown promise in improving feature extraction and scene understanding, their **integration into lightweight visual SLAM frameworks is still evolving**.
- **Therefore**, there is a research opportunity to develop a **low-cost, efficient visual SLAM framework using a monochromatic camera**, while incorporating **improved feature extraction techniques** to enhance robustness and localization accuracy.

Research Objectives

The objectives of this project are:

- **Develop a monocular SLAM system using a monochromatic camera** for localization and mapping.
- Implement a **feature-based visual odometry pipeline** for estimating camera motion.
- Construct a **sparse 3D map of the environment** using triangulation.
- Implement **loop closure detection** to reduce accumulated drift in long trajectories.
- Evaluate the system in terms of:
  - localization accuracy
  - mapping quality
  - computational efficiency.

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- https://www.sciencedirect.com/science/article/abs/pii/S1051200422001622?via%3Dihub
- https://arxiv.org/pdf/2305.04298
- https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2024.1424883/full
- https://www.nature.com/articles/s41598-025-91857-5.pdf
- https://github.com/isl-org/VI-Depth

Monochromatic Camera

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Image Preprocessing

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Feature Detection (ORB / FAST)

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Feature Matching

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Pose Estimation

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Triangulation

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Sparse Map Generation

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Loop Closure Detection

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Pose Graph Optimization